

A DGS-Based Symmetrical Patch THz Antenna for IoT and Future 6G Applications

Sayed Shifat Ahmed
Electrical and Electronic
Engineering

Rajshahi University of Engineering
and Technology
Rajshahi 6204, Bangladesh
shikor99@gmail.com

Shah Md Toufik Ahmed
Electrical and Electronic
Engineering

Rajshahi Science & Technology
University
Natore 6260, Bangladesh
toufik14.eece@gmail.com

Md. Mehedi Hassan
Information and Communication
Engineering

Technische Universität Darmstadt
64289 Darmstadt, Germany
hassan.eee18@gmail.com

Md. Arafath Hossain Rafi
Electrical and Electronic
Engineering

East West University
Dhaka 1212, Bangladesh
arafathrafi140790@gmail.com

Tithi Rani
Electronics and Telecommunication
Engineering

Rajshahi University of Engineering
and Technology
Rajshahi 6204, Bangladesh
tithi.saha.134040@gmail.com

Liton Chandra Paul
Electrical, Electronic and
Communication Engineering

Pabna University of Science and
Technology
Pabna 6600, Bangladesh
litonpaulete@gmail.com

Abstract—This manuscript delineates the conception of a diminutive, featherweight terahertz microstrip patch antenna, quintessential for IoT and 6G communication. The design methodology incorporated Rogers RT5880 as a substrate. The antenna's performance manifested a superior reflection coefficient of <-27 dB at two resonance frequencies. The expansive bandwidth of the antenna is attributed to its defected ground surface, achieving an impedance bandwidth of 2550 GHz, spanning from 2.95–5.50 THz. The antenna also exhibited a high gain and maximum radiation efficiency of 86%. A favorable VSWR value serves as a good impedance matching, reflecting optimal alignment between the transmission line and the load. The results demonstrate that the designed antenna radiates efficiently and is apt for future generation communication.

Keywords—Terahertz applications, 6G, IoT, symmetrical patch, Rogers RT5880, wideband, DGS.

I. INTRODUCTION

Terahertz communication spanning the 0.1–10 THz spectrum, promises ultra-high data rates and reliable connections for IoT, sensing, and security applications. Expected to play a significant role in 6G by 2030, terahertz (THz) antennas enable efficient signal transmission and ultra-high-capacity wireless backhaul, bridging connectivity gaps. This technology addresses demands for faster, lower-latency networks while unlocking opportunities in advanced wireless systems and contributes to a pivotal role in signals within this spectrum [1]. The primary technologies in 6G wireless communication networks emphasize THz channel modeling, designing multibeam antenna, development of front-end chips, baseband signal processing, and effective resource management [2]. The short wavelength of THz waves increases image resolution than microwave bands and facilitate creative solutions like wireless networks on chips (SoC) and the Internet of Nano-Things [IoNT] [3]. However, the shift of carrier frequency into the THz regime improve bandwidth and hence data speed (Tbps) but also imposes significant challenges due to the unique propagation characteristics of these frequencies, such as high path loss,

atmospheric absorption and manufacturing problems due to its miniaturized sizes [4]. Furthermore, the short wavelength of terahertz waves allows for the design of small and compact antennas which is beneficial for the miniaturization of devices [5]. The immense amount of data initiated by these devices necessitates a communication infrastructure capable of handling high data rates, which is where THz antennas come into play. Even though 5G technology is not fully deployed yet, researchers are already developing antennas for communication beyond 5G and 6G, which are expected to support all Internet of Things (IoT) services, satellite communications, as well as advancement in artificial intelligence (AI) and machine learning (ML) [3]. The future of wireless communication is exciting with THz antennas paving the way for a new era of ultra-fast, reliable, and intelligent communication systems with the ability to accommodate a high volume of connected devices, which makes it well-suited for IoT applications. However, deploying terahertz communication in 6G presents significant technical challenges that must be overcome consisting of high data rates and scarce commercial parts, necessitating creative strategies to address these challenges. To achieve this dependable 6G wireless communication, it is essential to develop and manufacture a high-gain terahertz antenna for both transmitters and receivers [6].

As advancements and innovations in this area progress, we can expect to see significant advancements in IoT and 6G applications. For these circumstances in [7], a novel $2000 \times 1000 \times 600 \mu\text{m}^3$ gold-based 1×4 antenna array for terahertz communications, fabricated using electron beam lithography. The array achieved a 400 GHz impedance bandwidth at a center frequency of 0.925 THz, offering high gain and minimal back lobe levels. The work in [8], a simulation of a terahertz microstrip patch antenna was suggested and utilizing Rogers RO3010 substrate for applications in 6G technology. The antenna resonates at a frequency of 0.854 THz, features a return loss of -19.5 dB, offers a limited bandwidth of 44 GHz, and has a gain of 3.88 dBi, demonstrating good radiation properties. On the other side M. Zubair et al. presented a truncated square patch on a polyamide substrate fed by a microstrip line [9], enhances

bandwidth to 410 GHz than [8], radiation efficiency to 80.2%, gain to 5.01 dBi, and directivity to 6.06 dBi at 3.65 THz. Document [10] by K. Vinodan et al. outlines a design of a microstrip patch antenna design for wearable medical imaging applications, functioning at a frequency of 0.775 THz, featuring a 20% impedance bandwidth and a gain of 7 dB. This paper discusses a model of $300 \times 300 \times 45 \mu\text{m}^3$ square slotted microstrip patch antenna supporting THz communication made from polyimide substrate and gold metal [11]. This design is optimized using HFSS simulator offers high gain and wide bandwidth (580-660 GHz). The graphene-based meta-surface patch antenna in [12] demonstrates quad-band functionality, resonating at frequencies of 1.762 THz, 2.134 THz, 2.462 THz and 2.782 THz achieving a peak gain of 5.37 dBi at 2.782 THz along with a fractional bandwidth of 23% at 1.762 THz. This makes it suitable for applications in spectroscopy, security scanning, and various medical uses within the terahertz spectrum. Another work [13] introduces a $1000 \times 1000 \times 100 \mu\text{m}^3$ sub-terahertz (THz) band microstrip omnidirectional antenna featuring a straightforward un-slotted design with an impressive average efficiency of 92% and stated possesses a broad impedance bandwidth of 0.114 THz similar design adapted into 2×2 MIMO antenna for increase gain of two times single element design and shows good isolation. A modified pill-shaped patch antenna on a FR-4 substrate operates at four distinct frequencies of ≈ 0.5 , ≈ 0.91 , ≈ 1.29 , ≈ 1.6 , ≈ 1.8 THz within 0.24 THz to 2.1 THz regions showing radiation efficiency of 87% for various THz applications presented in [14]. The document [15] by W. A. Khan et al. demonstrated a wideband THz micro-antenna on a silver substrate that attains a significant gain of 5.87 dB and over 80% radiation efficacy using a partial ground plane, a coplanar design, and two L-shaped stubs. The antenna, simulated using CST Microwave Studio, reports a bandwidth of 0.415 THz. Two microstrip antennas with S-shape and Z-shape slots were studied in [16], each measuring $1200 \times 460 \times 127 \mu\text{m}^3$ and showing left-hand and right-hand circular polarization, respectively. Designed for THz wireless communication, they operate within 0.258 - 0.355 THz, achieve a max gain of 5.6 dBi at 0.281 THz, and have a 93% radiation efficiency. The authors, M. S. Rabbani et al. designed three THz microstrip patch antenna arrays at 0.835 THz, 0.635 THz, and 0.1 THz frequencies on Liquid Crystalline Polymer (LCP) substrate for medical applications in [17]. Only the 0.1 THz array ($2489 \times 3078 \times 100 \mu\text{m}^3$) was fabricated and tested, showing a minimum S11 of -26.04 dB at 100.5 GHz and a bandwidth of 2.24 GHz.

Considering the limitations of existing studies and to address the previously discussed challenges, this study introduces a high-gain slotted dual circular-shaped a microstrip patch antenna designed for THz applications is presented. This antenna boasts a symmetrical design and provides an impressive impedance bandwidth of 2550 GHz. Furthermore, the integration of a defected ground surface (DGS) significantly enhances the antenna gain (6.55 dB). The design process and geometric considerations are detailed in section II, while the parameters and radiation characteristics are presented in section III and finally the proposed model is compared with previous works and conclusion. This comparative analysis unequivocally validates the aptness of the proposed antenna, thereby establishing its potential for integration into forthcoming wireless communication applications.

II. ANTENNA DESIGN

The geometrical dimension and optimized parametric values of the proposed antenna are depicted in Fig. 1 and Table-I respectively. The antenna under proposal, designed to function within the THz spectrum, necessitates minimal dimensions in terms of both width and length. Consequently, it requires the thinnest possible substrate. Rogers RT5880 has been chosen as the substrate material ($\epsilon_r=2.2$, $\tan\delta=0.0009$), featuring a thickness ranging from 13 to 39 μm . The design proposed encompasses an antenna of $215 \times 220 \mu\text{m}^2$, operating within the THz frequency range. Fig. 1 represents microstrip patch front view consisting of a resonating patch made from copper, which serves as the source of radiant energy. It is supported by a substrate composed of dielectric material that helps guide the transverse electric field. Additionally, the ground layer (Fig. 1b) functions as a reflector, bouncing back the radiation that is directed towards the opposite side of the antenna. The conductive patch and the defected ground layer are made from copper with a thickness of 2 μm , as shown in Fig. 1(a, b). The antenna design is square-shaped, and the radiating patches, which have a duo-circular design, are positioned above the substrate, illustrated in Fig. 1(c). The antenna's design is heavily influenced by the design of all these components. A 30 μm microstrip feed line is utilized to energize the antenna. The design is modeled, simulated, and optimized using a CST simulator. The antenna dimensions listed in Table I are obtained using traditional antenna theory equations outlined below [18]:

The resonant frequency of a circular patch antenna is given by,

$$f_r = \frac{c}{2\pi a} \sqrt{\frac{\epsilon_r + 1}{2}} J'_{0m} \left(\frac{2\pi a}{\lambda_0} \sqrt{\frac{\epsilon_r + 1}{2}} \right) \dots \dots \dots (1)$$

where c is the speed of light, a is the radius of the patch, ϵ_r is the relative dielectric constant of the substrate, J'_{0m} is the derivative of the zeroth-order Bessel function of the first kind, and λ_0 is the free-space wavelength.

The effective dielectric constant of a circular patch antenna is given by,

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{12h}{\pi a} \right)^{-\frac{1}{2}} \dots \dots \dots (2)$$

where h is the substrate thickness, and a is the radius of the patch.

The radius of a circular patch antenna is given by,

$$a = \frac{c}{2\pi f_r} \sqrt{\frac{\epsilon_r + 1}{2}} J'^{-1}_{0m} \left(\frac{2\pi f_r}{c} \sqrt{\frac{\epsilon_r + 1}{2}} \right) \dots \dots \dots (3)$$

where c is the speed of light, f_r is the resonant frequency, ϵ_r is the relative dielectric constant of the substrate, and J'^{-1}_{0m} is the inverse function of the derivative of the zeroth-order Bessel function of the first kind.

Therefore, the equations for the substrate length (L_g) and width (W_g) are:

$$L_g = L_p + 6h\sqrt{\epsilon_r} \dots \dots \dots (4)$$

$$W_g = W_p + 6h\sqrt{\epsilon_r} \dots \dots \dots (5)$$

where L_p and W_p are the patch length and width, h is the substrate height and ϵ_r is the relative dielectric constant of the substrate.

While the values derived from the preceding equations may not consistently represent the optimal solutions, the precise values are ascertained by adhering to the workflow delineated in Fig. 2. Initially, a conventional microstrip

antenna with copper patch was studied. Then a simple microstrip fed patch antenna is designed with intersection of two circular shaped patches with covered full ground surface as illustrated in Fig. 2 stage-I which resonant at 1.85 THz and comes very narrow bandwidth. Creating inner circle within the previous circles shows good gain but doesn't come desired result in stage-II. The main goal of this research is to develop a compact and efficient antenna that demonstrates both wide bandwidth and high gain for THz applications. To achieve these desired characteristics, several approaches will be explored, we further optimized the design by etching some portion in the annulus of circles stage-III and finally modified ground plane by dividing it two portions Fig. 2, stage-IV which shifts a wideband frequency (2550 GHz) ranging from 2.95-5.5 THz and making it two resonances point between the entire bandwidth. The complete design evolution of the proposed antenna, along with the corresponding reflection coefficients, is presented in Fig. 3.

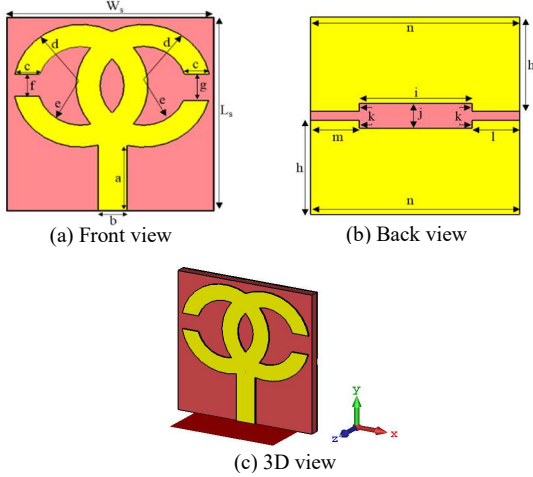


Fig. 1. DGS-based symmetrical patch THz antenna.

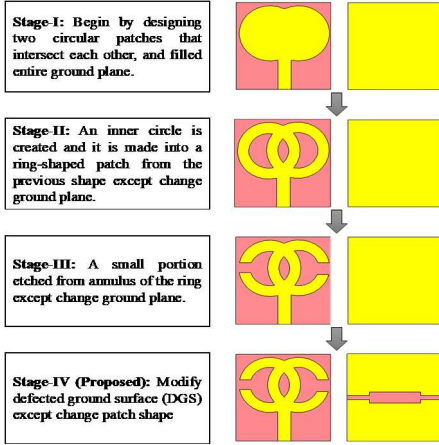


Fig. 2. Design flow of proposed THz antenna.

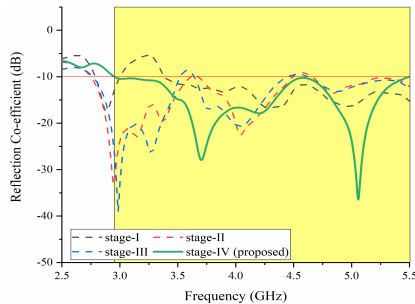


Fig. 3. Reflection co-efficient responses of each design stage of the proposed antenna.

TABLE I PHYSICAL DIMENSIONS OF PROPOSED ANTENNA

Parameter	Value (μm)	Parameter	Value (μm)
Substrate width (W_s)	215.00	Right cutting edge (g)	30.01
Substrate length (L_s)	220.00	Ground plane width (h)	105.00
Feeder length (a)	74.10	Central ground (i)	116.00
Feeder width (b)	30.00	Ground plane spacing (j)	28.00
Annulus width (c)	25.70	Cavity height (k)	9.00
Outer circle radius (d)	70.00	Right ground (l)	49.00
Inner circle radius (e)	45.00	Left edge ground (m)	50.00
Left cutting edge (f)	25.00	Ground plane length (n)	215.00

III. RESULTS AND PARAMETRIC INVESTIGATION

The performance of the designed antenna is assessed in free space, focusing on essential factors such as impedance matching, reflection coefficient, efficiency, Voltage Standing Wave Ratio (VSWR), and the distribution of current throughout the antenna. Furthermore, the evaluation includes the polar plot radiation pattern, results from three-dimensional radiation patterns, directivity, and the gain attained by the antenna at its resonant frequency.

A. S-parameter

The performance of an antenna is typically categorized into two broad parameters: input and output. The input parameters include the return loss (S_{11}), Voltage Standing Wave Ratio (VSWR) and input impedance. Therefore in Fig. 4, the designed DGS THz antenna structures demonstrate simulated reflection coefficients that are significantly lower than -10 dB at their central resonance frequencies of 3.7 THz and 5.06 THz, measuring -27.98 dB and -36.38 dB respectively. This implies that the antenna is designed to operate efficiently at these specific frequencies in the terahertz (THz) range and a very small fraction of the incident power is reflected back at these frequencies. This suggests that the antenna structures are well-matched to the transmission line at these points, leading to maximum power transfer. As observed from Fig. 4, the suggested THz antenna functions within a frequency range of 2.95 to 5.5 THz, featuring an operating bandwidth of 2550 GHz. This antenna structure is designed to efficiently resonate at multiple frequencies (e.g., 3.7 THz and 5.06 THz), enabling effective radiation rather than reflection losses. The presence of multiple resonant modes enhances its versatility, allowing integration into different communication systems operating across various bands. The CST simulation yielded VSWR values of 1.08 and 1.03, corresponding to resonance frequencies and is represented by Fig. 5. The effectiveness of an antenna can be negatively affected by the presence of standing wave voltage in the feed line, which occurs when a signal reflects to the source. This reflection arises from an impedance mismatch between the source and the antenna. In an ideal case, the VSWR value would be 1, signifying that the antenna accepts 100% of the power with no reflection. The proposed terahertz antenna demonstrates broadband performance with a stable and acceptable VSWR of 1–2, as shown in Fig. 5. This value is maintained across a frequency band that spans from 2.95 to 5.5 THz. The degree of impedance matching between the input signal and the feedline is indicated by the input impedance. When the impedance aligns closely with 50 ohms, it is considered a perfect impedance matching. Fig. 6 shows the proposed

antenna resonating at 3.7 THz and 5.06 THz with input impedances of 59 Ω and 58.8 Ω , despite the challenge of multiband matching across the -10 dB bandwidth.

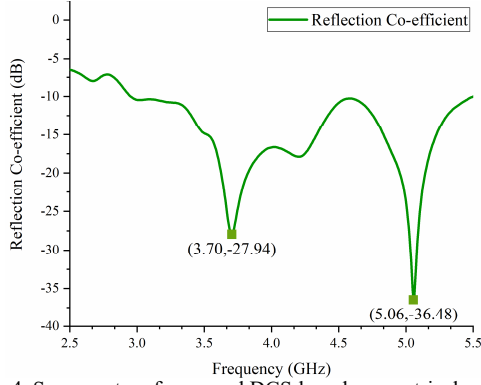


Fig. 4. S-parameter of proposed DGS-based symmetrical patch THz antenna.

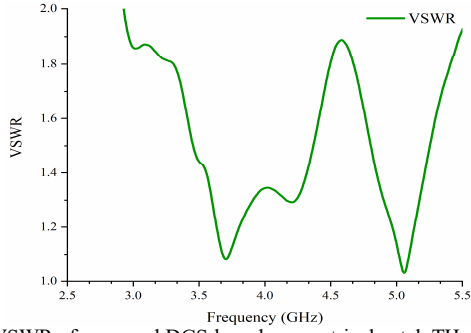


Fig. 5. VSWR of proposed DGS-based symmetrical patch THz antenna.

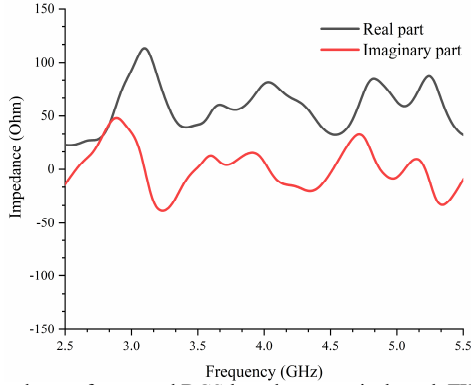


Fig. 6. Impedance of proposed DGS-based symmetrical patch THz antenna.

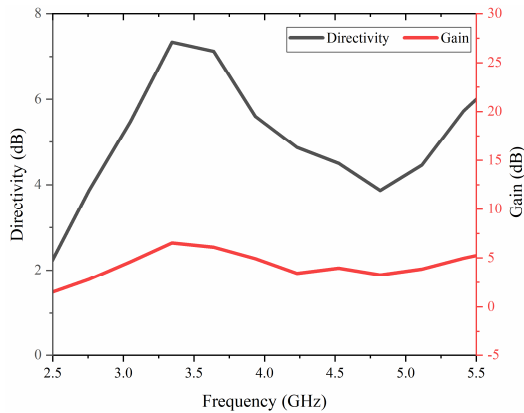


Fig. 7. Gain and directivity of DGS-based symmetrical patch THz antenna.

B. Gain, Directivity and Radiation Efficiency

Resonant gain is essential, as return loss and VSWR alone do not guarantee radiation efficiency; accordingly, Fig. 7 presents the proposed gain of antenna across all operating

frequencies, showing positive values from 3.50 to 5.06 dB and at the resonant frequencies of 3.7 THz and 5.06 THz corresponding gains are 5.86 dB and 3.75 dB, respectively, while the maximum gain of 6.55 dB is observed at 3.35 THz.. This high-quality gain value indicates the effectiveness of this micro-sized antenna. Hence, the gain of the proposed antenna design plays a pivotal role in its overall performance evaluation. Fig. 7 aligning with the gain plot also depicts the directional performance of the antenna. Radiator needs to point in the right direction for optimal functionality otherwise applicability could be restricted within the relevant frequency band. The plot shows directivity of 6.81 dB, 4.35 dB at resonance frequencies. Whether plot of in Fig. 8 shows the variation in efficiency with respect to frequency. As the frequency increases, efficiency experiences a reduction because of ohmic and radiation losses. The system achieves a peak radiation efficiency of 87%. An examination of the efficiency plot reveals that the system consistently maintains an efficiency between 80-87% across the entire operating frequency range.

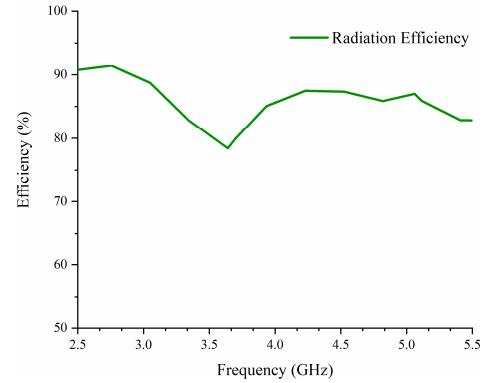


Fig. 8. Radiation efficiency of DGS-based symmetrical patch THz antenna.

C. Radiation Pattern

The two-dimensional radiation patterns, recorded at two different resonant frequencies of 0.96 THz and 1.28 THz, are illustrated in Fig. 9. In $\phi=90^\circ$, the electric (E) and magnetic (H) field shows maximum half power beamwidth (HPBW) of 23.6 at an angle of 42° when it exhibits lowest sidelobe of -2.3 dB (max). At $\phi=90^\circ$, 0° the main lobe magnitudes for the E and H fields are 20.7 dBV/m and -32.4 dBA/m, respectively (Fig. 9a). Conversely, at the resonance frequency of 5.06 THz, the major lobe of the electric field radiates a maximum energy of 23.5 dBV/m at an angle of 120° and magnetic field radiates -33.7 dBA/m peak. The lowest side lobes of -2.4 dB remain consistent under any condition of $\phi=0^\circ, 90^\circ$ for both E and H fields. However, in Fig. 9(b) the HPBW reaches a maximum of 30.1 for both E and H fields when $\phi=0^\circ$. Fig. 10 illustrates the 3D gain radiated in the azimuthal plane, as represented by the polar plot.

D. Surface Current Distribution

The electromagnetic behavior of the patch antenna along with the current surface density at resonance frequencies are visualized in Fig. 11. Here a defective ground structure (DGS) based circular ring symmetrical patch terahertz (THz) antenna with line feeding is simulated using CST Microwave Studio. It can be observed in Fig. 11 (a) that the surface current density is predominantly higher in the lower region of the patch, specifically at the feed point but in Fig. 11 (b) most current also concentrate vertically aligned to the middle of the

substrate. The intersection of the circular ring and the ground plane plays a significant role in this distribution.

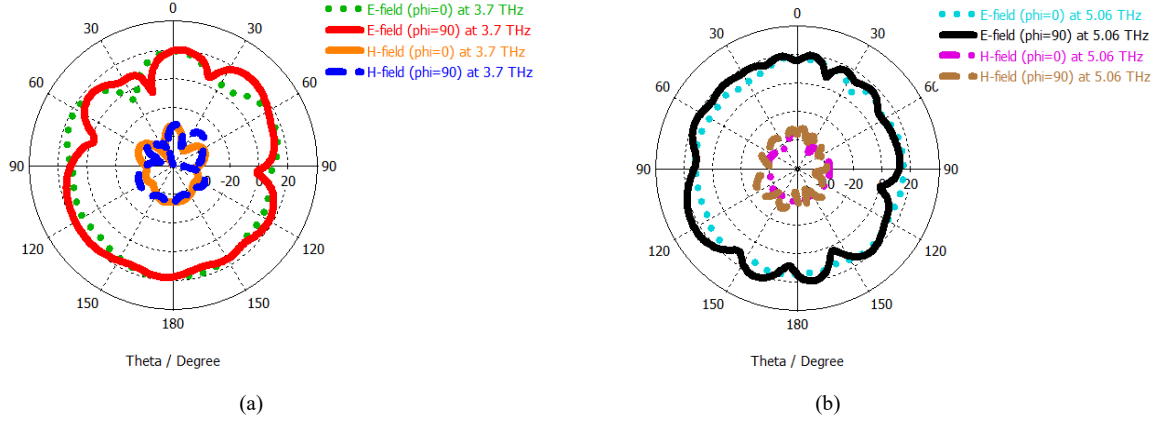


Fig. 9. E, H field analysis of proposed DGS-based symmetrical patch THz antenna at (a) 3.7 THz and (b) 5.06 THz.

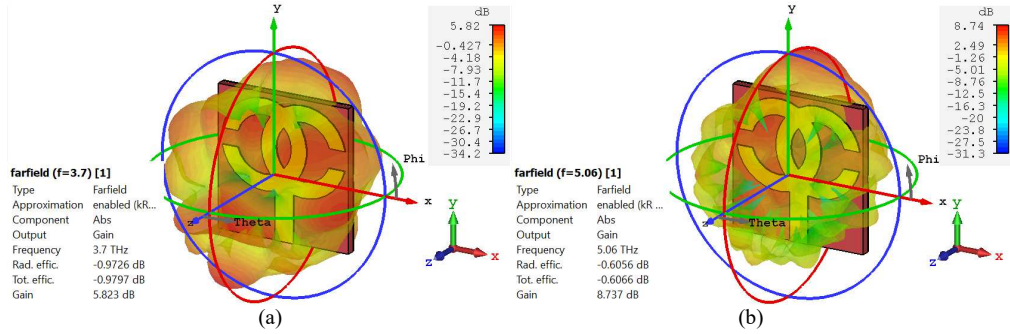


Fig. 10. 3D radiation pattern of proposed DGS-based symmetrical patch THz antenna at (a) 3.7 THz and (b) 5.06 THz resonance frequency.

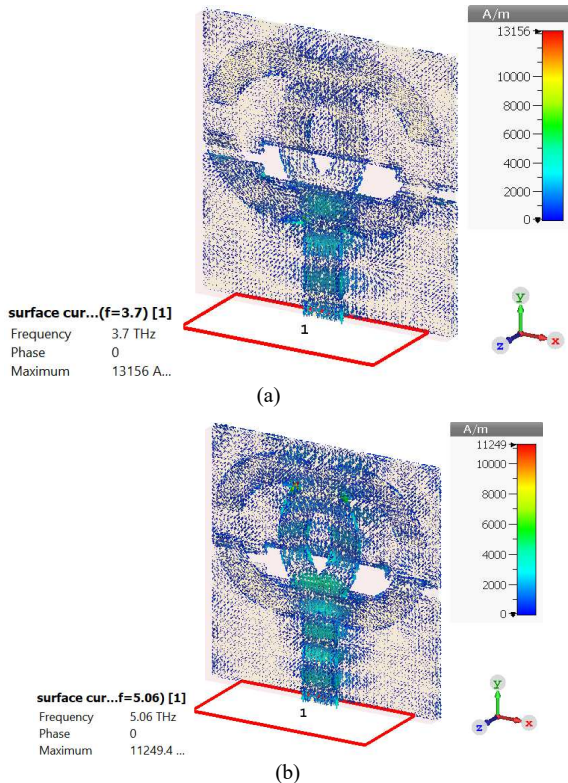


Fig. 11. Current distribution at (a) 3.7 THz and (b) 5.06 THz resonance frequencies.

IV. COMPARATIVE ANALYSIS

For instance, a comparative study in literature theoretically analyses in section I. Table-II delineates the

performance disparities between the proposed antenna and other comparable terahertz antennas. It's noteworthy that certain parameters are absent in various studies (VSWR, directivity). However, cost considerations are critical and the design objective is to ensure that the antenna exhibits optimized performance across all aspects, balancing both efficiency and cost.

V. CONCLUSION AND DISCUSSION

In conclusion, the paper details the design and evaluation of a compact and lightweight THz microstrip patch antenna, which is ideal for integration into compact electronic devices and is intended for use in IoT and 6G communication. The design process, executed in four iterations, involved the use of Rogers RT5880 as a substrate and copper for the radiating patches, with each step performance tested as it was added. This work simulated using CST Microwave Studio and comparing design performance with others in Table II, showcasing the proposed symmetrical patch antenna attained an impressive reflection coefficient of around <-28 dB at resonance frequencies of 3.7 and 5.06 THz. Its DGS structure contributes to its broad bandwidth and achieves a substantial impedance bandwidth (2550 GHz) ranging from 2.95-5.5 THz. The antenna also demonstrated a peak gain of 6.55 dB, a peak directivity of 6.81 dB and a radiation efficiency of 86% maximum. A favorable VSWR value (1.08) further indicates good impedance matching. Upon comparison with previous work, the antenna that has been designed radiates effectively and is suitable for future generation communication. This analysis highlights the potential of THz antenna for advancing IoT and 6G communication technologies for future generations.

TABLE II COMPARISON TABLE

Ref.	Size (L×W×h) μm ³	Substrate material	Patch material	Operating frequency range (THz)	BW (GHz)	Reflection coefficient (dB)	Centre frequency (THz)	Max Gain (dB)	Max Efficiency (%)	Application
[7]	2000×1000×600	Roger RT5880	Gold	0.75-1.10	350	≈-48	0.925	12.5- 16	85	Terahertz Communications
[8]	200×200×50	Rogers RO3010	Copper	≈0.861- 0.905	44	-19.5	0.854	3.88	NA	6G Future Communications
[9]	100×100×10	Polyamide	-	3.48-3.90	410	-29	3.65	5.01	80.2	Terahertz Communication
[10]	280×242×25	Polyamide	Gold	≈0.73-0.81	≈80	-24	0.775	7	NA	Medical Imaging Application
[11]	300×300×45	Polyamide	Gold	0.58-0.66	80	-25	0.625	7.155	NA	Terahertz Wireless Communication
[15]	300×300×45	Polyamide	Silver	0.523- 0.9395	415	≈-16, ≈-14.2	≈0.645, ≈0.87	5.87	>80	6G Application
[16]	1200×460×127	RT6006	Copper	0.258- 0.355	97	≈-15	0.3	5.6	93	THz Communication
[17]	2489×3078×100	Liquid Crystalline Polymer	Copper	≈0.09925- 0.10105	1.8	≈-28, ≈-40	≈0.624, ≈0.634	15.82	52	Medical Applications
Proposed work	215×220×13.39	Rogers RT5880	Copper	2.95-5.5	2550	-27.98, -36.38	3.70, 5.06	6.55	86	IoT and Future 6G Applications

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